

platform chooses a sharing network with minimal homophily to maximize the spread and appeal of the content throughout the user community. In contrast to this case, when the relevant articles have lower reliability, the platform chooses a network with maximal homophily and recommends articles to users with aligned beliefs. These articles then spread rapidly in the “filter bubble” the platform’s algorithms have created—because now ideologically like-minded individuals know that they are unlikely to be caught sharing misinformation in their extreme echo chambers.

We also study regulations aimed at minimizing the spread of misinformation. Revealing the provenance of a news item (for example, providing full context for a quote) can be useful, because this additional information allows users to more easily fact-check the content for veracity. However, we also show that this type of regulation may backfire because of a “bystander effect” whereby future users assume content has been previously fact-checked and put more faith in it. Similarly, tagging or censoring articles from less reliable sources may make agents more confident in those that remain on the site untagged—thus creating a Bayesian version of the “false sense of security”. Finally, we show that regulating platform algorithms that shape sharing network properties (in particular, homophily) can be effective.

Our framework was purposefully chosen to be simple and several generalizations would be interesting to consider in future work. Most importantly, our assumption that agents are Bayesian rational should be viewed as a useful, albeit not fully realistic, benchmark. In our setting, it brought out certain new strategic forces—highlighting how social media actions exhibit strategic complementarity and how the degree of homophily alters agents’ strategic behavior. Nevertheless, misinformation may be particularly damaging when agents have cognitive limitations and are only boundedly rational. Incorporating such considerations is one of the most important directions for future research. Interesting questions that emerge in this case relate to whether the platform, in addition to designing algorithms that create filter bubbles, may also choose strategies that exploit the cognitive limitations of users.

Other theoretical generalizations that might be interesting to consider include extensions to repeated interactions with incomplete information, which would enable agents to also update their beliefs about the ideological position of other agents in their sharing network. Fully endogenizing reputational concerns, taking into account the network position of agents, would be another interesting direction for future research. In this case, the existing reputational capital of an agent will determine how likely she is to risk sharing misinformation. We can also use this extended setup with repeated interactions to study how agents update their initial political views. When there is limited misinformation, agents will gradually learn the true state. In contrast, when there is a significant probability of misinformation, agents will be uncertain about how to interpret articles that disagree with their priors and this may place an upper bound on the speed and possibility of learning (see [Acemoglu et al. \(2016\)](#)).

Despite its simplicity, our model makes several new empirical predictions, most notably related to the non-monotonic effects of homophily and polarization and to platform incentives and algorithmic decisions. Investigating these predictions empirically as well as generating new stylized facts about patterns of these information cascades on social media, is another important area for future research.